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DS PROGRAMMING

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CO5-AT2-Short Answers Test

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QUESTIONS

Q1
What is the time complexity of Prim's algorithm using an adjacency matrix?
2 marks

Q2
Define the Floyd–Warshall algorithm.
2 marks

Q3
What is Breadth First Search (BFS)?
2 marks

Q4
Which data structure is used in BFS?
2 marks

Q5
What is Depth First Search (DFS)?

Q1 What is the time complexity of Prim's algorithm using an adjacency matrix? 2 marks

What is the time complexity of Prim's algorithm using an adjacency matrix?

Your Answer

O(V^2), where V is the number of vertices.

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What is Depth First Search (DFS)?

Q2 Define the Floyd–Warshall algorithm. 2 marks

Define the Floyd–Warshall algorithm.

Your Answer

The Floyd–Warshall algorithm is a dynamic programming algorithm used to find the shortest paths between all pairs of vertices in a weighted graph, with a time complexity of O(V^3).

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2 marks

Q3 What is Breadth First Search (BFS)? 2 marks

What is Breadth First Search (BFS)?

Your Answer

Breadth First Search (BFS) is a graph traversal algorithm that visits all neighbor nodes at the present depth level before moving on to the nodes at the next depth level.

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Q4 **Which data structure is used in BFS?** 2 marks

Which data structure is used in BFS?

Your Answer

Queue

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Q2
Define the Floyd-Warshall algorithm.
2 marks

Q3
What is Breadth First Search (BFS)?
2 marks

Q4
Which data structure is used in BFS?
2 marks

Q5
What is Depth First Search (DFS)?
2 marks

Q6
Which data structure is commonly used in DFS?
2 marks

Q7
A graph contains the vertices A, B, C, D, E, F with edges A-B, A-C, B-D, B-E, C-F, E-F. Starting from A, determine the BFS

DFS is a graph/tree traversal method that explores one branch as far as possible, then backtracks; it uses a stack (or recursion).

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Q5 What is Depth First Search (DFS)?

Q6 **Which data structure is commonly used in DFS?** 2 marks

Which data structure is commonly used in DFS?

Your Answer

A stack is the data structure commonly used in Depth First Search (DFS).

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Q7 **A graph contains the vertices A, B, C, D, E, F with edges A-B, A-C, B-D, B-E, C-F, E-F. Starting from A, determine the BFS traversal order. A social network represents users as ve** 2 marks

A graph contains the vertices A, B, C, D, E, F with edges A-B, A-C, B-D, B-E, C-F, E-F. Starting from A, determine the BFS traversal order.
A social network represents users as ve

Your Answer

A, B, C, D, E, F

Save Answer